

PHC Pulse — Database Schema & Data Dictionary

Filled · WHAT DATA does the system use?

Purpose. Define the data contract so the coding agent does not invent incompatible tables or fields.

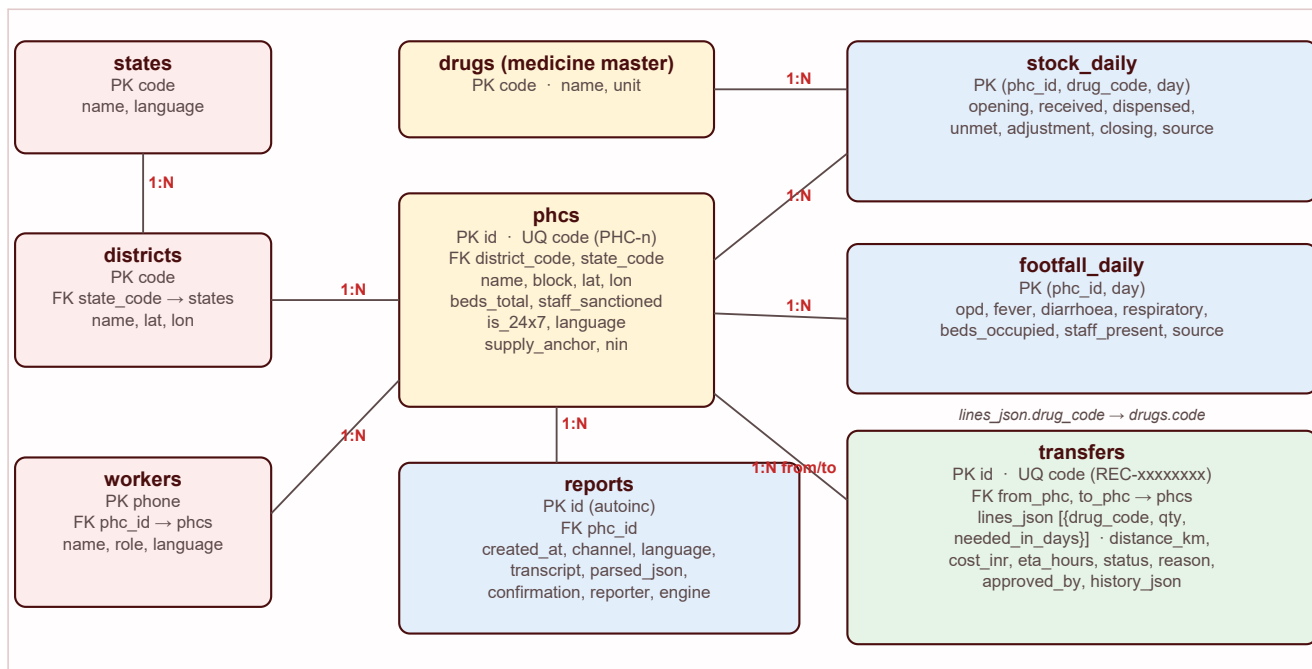
Filled from the repository *PHC-Pulse-GDG-Code-for-community* (main @ fd42d41) on 26 Sep 2026. Numbers quoted were measured by running the engine and the 13-test pytest suite (all passing) with *PHC_TODAY=2026-09-25*, seed 42.

Status legend: **[IMPLEMENTED]** in the repo and tested **[PARTIAL]** partly built **[PLANNED]** designed here, not yet in code

1. Dataset Inventory

Dataset	Source	Real/Synthetic	Purpose
PHCs	backend/app/reference.py + seed.build_phcs: 4 states, 8 districts × 15 real block names; 10- or 6-bed PHCs, 9–14 sanctioned staff, HMIS NIN	Semi-real: real district centroids and block names; PHC coordinates jittered; synthetic NIN	Facility master, map, scoping, distances, capacity
Inventory	seed.simulate → <i>stock_daily</i> (180 days × 120 PHCs × 10 NLEM drugs). Live rows from voice/USSD/SMS/IVR reports and transfers	Synthetic history + live demo input	Current stock, stock-out labels, surplus, ledger for transfers, HMIS export
Daily demand	Derived: dispensed + unmet per PHC × drug × day (footfall × drug rate × state epidemiology factor, Poisson)	Synthetic	Forecaster training, backtest, BQML / Vertex training view
Footfall	seed.simulate → <i>footfall_daily</i> : OPD, fever, diarrhoea, respiratory, beds occupied, staff present (weekday + monsoon seasonality, 3 injected outbreaks)	Synthetic, sized to HMIS PHC OPD ranges	Anomaly / cluster detection, surge factor, resilience score, capacity
Transfers	Created in-app when an officer approves a MILP recommendation; lifecycle updates	Generated live (starts empty)	Pipeline stock (incoming/committed), audit trail, acceptance metric

2. ER Diagram



Source of truth: *backend/app/db.py* (SQLite). *bigquery/schema.sql* mirrors it and adds *state_code* to fact tables, *DAY* partitioning, clustering (*state_code*, *phc_id*, *drug_code*) and row access policies.

3. Tables / Collections

Template names are mapped to the repo's actual tables so a coding agent does not invent parallel ones.

Table	Primary key	Fields	Relationships
phcs	id INTEGER (code 'PHC-n' UNIQUE)	name, block, district_code, state_code, lat, lon, beds_total, staff_sanctioned, is_24x7, language, supply_anchor (date of a monthly indent), nin (HMIS NIN)	N:1 districts, states; 1:N stock_daily, footfall_daily, workers, reports; 1:N transfers (from/to)
medicine_master → drugs	code TEXT (PCM, ORS, ZNC, AMX, IFA, MET, AML, ACT, ASV, OXY)	name, unit, reference.DRUGS also holds rate/patient, driving syndrome, local names, aliases and SHORTAGE_PENALTY (INR)	1:N stock_daily; referenced by transfers.lines_json
inventory → stock_daily	(phc_id, drug_code, day)	opening, received, dispensed, unmet, adjustment, closing, source (hmis voice ussd sms ivr transfer)	N:1 phcs, drugs. Closing = opening + received – dispensed + adjustment
daily_demand → view	(phc_id, drug_code, day)	demand = dispensed + IFNULL(unmet,0) WHERE dispensed IS NOT NULL (engine.py, forecast.sql, Vertex training view)	Derived from stock_daily
footfall → footfall_daily	(phc_id, day)	opd, fever, diarrhoea, respiratory, beds_occupied, staff_present, source	N:1 phcs; joins stock_daily on (phc_id, day) for training
transfers	id INTEGER (code = recommendation id, UNIQUE)	from_phc, to_phc, lines_json, distance_km, cost_inr, eta_hours, status (approved in_transit delivered cancelled), reason, created_at, updated_at, approved_by, history_json	N:1 phcs twice. Status moves write stock_daily adjustments/receipts
scenario_truth	(not a table) scenario id	[PARTIAL] Ground truth lives in seed.py pins (PHC-14/22, KRP diarrhoea, GKP fever, lone KLB respiratory) and in the test assertions. Planned: a <i>scenario_truth</i> (scenario_id, phc_id, drug_code/syndrome, expected_status, expected_value) table for scoring.	References phcs, drugs
states / districts / workers / reports	code / code / phone / id	language; lat, lon; name, role, phc_id, language; channel, transcript, parsed_json, engine...	Master data, caller-ID lookup, report audit log

4. Field Dictionary

Field	Type	Required	Validation	Meaning
phc_id	string*	Yes	Unique	Facility id. *Repo: INTEGER phcs.id with a unique string code 'PHC-n' and the HMIS NIN. Foreign key in every fact table.
medicine_id	string	Yes	FK	Repo name drug_code . Must be in the 10-item NLEM catalogue; any other code (e.g. a Gemini hallucination) is dropped.
stock_quantity	integer	Yes	>=0	Repo closing (REAL, whole units in the drug's unit: strips, sachets, courses, vials, ampoules). Stock on shelf at end of day; reports clamp to ≥ 0.
expiry_date	date	No (planned)	Valid	[PLANNED] Not stored yet. Planned on stock_batches: ISO date > received date; drives the expiry-rescue rule (< 90 days).
patient_count	integer	No	>=0	Repo opd : outpatients seen today. Null when the PHC has not reported. Syndromic splits: fever, diarrhoea, respiratory.
quantity_used	number	No	>=0	Repo dispensed (+ unmet = true demand). Null on rows opened only by a transfer. On report: opening + received + adjustment – remaining.

5. Data Quality Rules

Relationships, missing values, ranges, dates, duplicates, synthetic-data labels

- **Relationships:** every fact row's phc_id exists in phcs and its drug_code in drugs. Unknown PHC → 404/400; unknown drug codes are skipped at ingest; USSD/IVR callers must exist in workers.
- **Missing values:** NULL means "not reported", never 0. Forecasting treats NaN as unobserved (state carried forward). A PHC that has not reported today loses 5 resilience points and uses yesterday's stock.
- **Ranges:** quantities ≥ 0. beds_occupied ≤ beds_total (clamped). staff_present ≤ staff_sanctioned in seed. Audio ≤ 8 MB. Surge factor clipped to 1–3×. Seasonal factors clipped to 0.05–5.
- **Dates:** ISO YYYY-MM-DD. One row per (phc, drug, day) and (phc, day). "Today" = PHC_TODAY or the system date. Report timestamps are local ISO seconds.
- **Duplicates:** composite primary keys make a second same-day report an update (INSERT OR REPLACE / UPDATE), not a new row. Transfers are unique by recommendation code, so a repeat approve returns the existing

transfer. USSD sessions submit once.

- **Synthetic label:** source='hmis' marks seeded backfill; live rows carry their channel. The UI footer and Models page state "Synthetic data modelled on HMIS". No patient-level data anywhere.

6. Sample Records

Taken from the seeded database (seed 42, PHC_TODAY=2026-09-25).

```
phcs      (14, 'PHC-14', 'PHC Anandapuram', 'Anandapuram', 'VPM', 'TN', 12.212, 79.612, 6, 13, 0, 'ta', '2026-04-25', '2000110866')
          (22, 'PHC-22', 'PHC Vandavasi', 'Vandavasi', 'TVM', 'TN', 12.498, 79.585, 10, 10, 1, 'ta', '2026-04-18', '2000174218')
drugs      ('PCM', 'Paracetamol 500mg', 'strips') ('ORS', 'ORS sachet', 'sachets')
stock_daily (22, 'PCM', '2026-09-25', opening 338, received 0, dispensed 22, unmet 0, adjustment -142, closing 174, 'hmis')
          (22, 'PCM', '2026-09-24', opening 369, received 0, dispensed 31, unmet 0, adjustment 0, closing 338, 'hmis')
footfall_daily (91, '2026-09-24', opd 52, fever 10, diarrhoea 17, respiratory 10, beds_occupied 6, staff_present 8, 'hmis')
workers      ('+919000000022', 'ANM Vandavasi', 'anm', 22, 'ta')
states      ('TN', 'Tamil Nadu', 'ta') districts ('TVM', 'TN', 'Tiruvannamalai', 12.23, 79.07)
reports      {phc_id:22, channel:'voice', language:'en', engine:'rules-fallback', reporter:'worker',
              parsed_json:{ "stock": [{"drug_code": "PCM", "quantity": 150, "kind": "remaining"},
                              {"drug_code": "ORS", "quantity": 60, "kind": "remaining"}], "beds_occupied": 3, "staff_present": 9, "opd_count": 72}}
transfers    {code: 'REC-C4290993', from_phc: 14, to_phc: 22, lines_json: [{"drug_code": "PCM", "qty": 284, "needed_in_days": 9}],
              distance_km: 41.5, cost_inr: 1347, eta_hours: 3.2, status: 'approved', approved_by: 'District Health Officer,
              Tiruvannamalai', reason: 'PHC-14 has surplus Paracetamol 500mg; PHC-22 needs it in 9 days',
              history_json: [{"status": "approved", "at": "...", "by": "District Health Officer, Tiruvannamalai"}]}
```